

Dynamics II, Forces Main Note

Effects of Forces on the motion of a body

- body at rest moves
- moving body increases speed
- Moving body decreases in speed
- moving body changes direction

When force applied on object $\Rightarrow$ there is change in speed/ and or direction $\Rightarrow$ there is a change in velocity
Thus there is acceleration (or deceleration)

- thus, Force $\Rightarrow$ can cause object to accelerate or decelerate

0 Acceleration $\Rightarrow$ implies an object can be stationary or moving with constant velocity.

$\Rightarrow$ even though acceleration = 0, it does not mean there are no forces acting on it. it just means that the resultant of these forces is 0

Newton's Laws

[Newton's First Law](#)

[Newton's Second Law](#)

[Newton's Third Law](#)

Inertia

[Inertia](#)

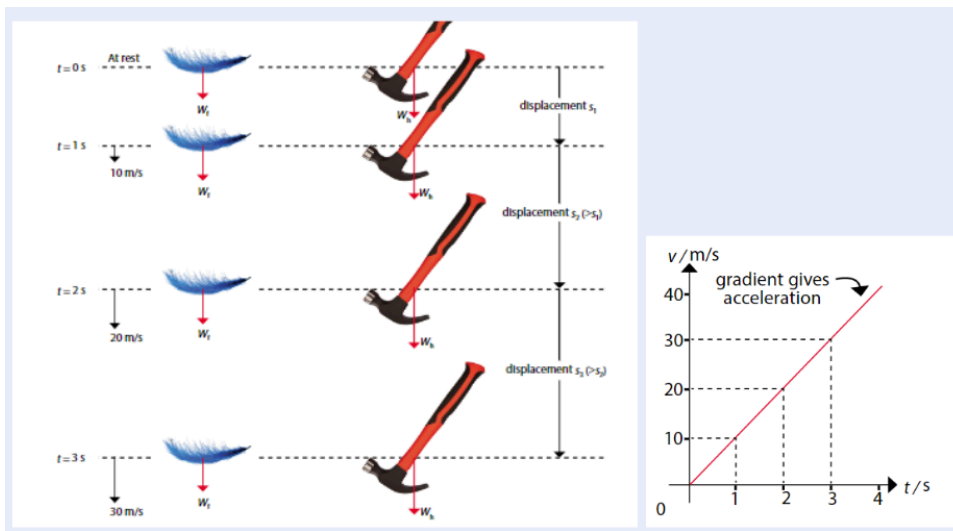
Some effects of Resistive Forces on Motion

[Friction and its Effects](#)

Objects Falling without Air Resistance

Objects can only be in free fall if the only force acting on it is its own weight

Vacuum



- Velocity of 2 objects under gravity increased by 20m/s every second (increases uniformly). both objects undergo constant acceleration of 10m/s^2
- Direction of their motion is downward (towards center of earth)
- Acceleration of free falling objects $\Rightarrow$ does not depend on their mass or size, all objects fall freely at a constant acceleration of 10m/s^2 (look at graph)

Objects Falling with Air Resistance

- always opposes the motion of moving objects
- increases with speed of the objects
- increases with surface area (or size) of the objects
- increases with density of the air

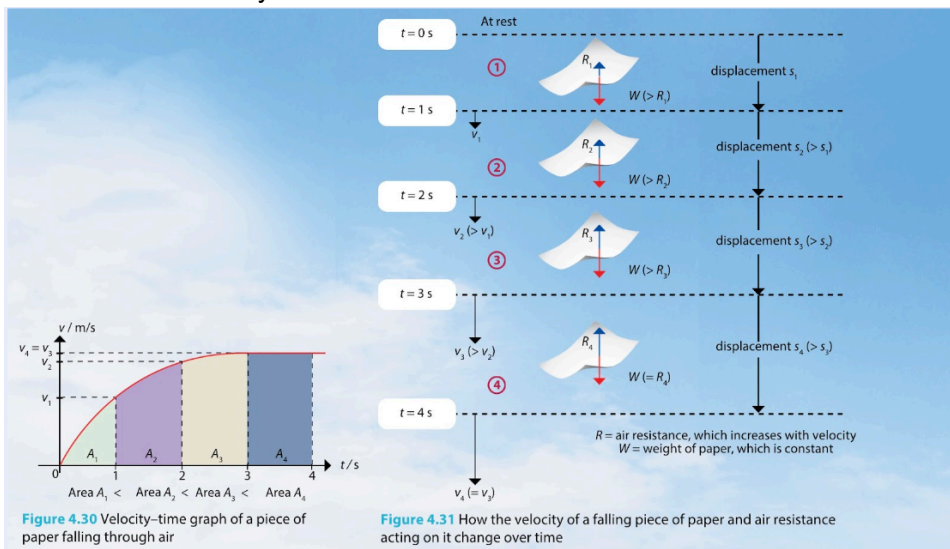


Figure 4.30 Velocity-time graph of a piece of paper falling through air

Figure 4.31 How the velocity of a falling piece of paper and air resistance acting on it change over time

Time Interval	Forces Involved	Changes in Velocity and Acceleration	Displacement
1	$W > R$	Velocity increases from zero to V_1 there is acceleration	$S_1 = A_1$
2	$W > R_2$ $R_2 > R_1$	velocity increases from V_1 to V_2 ΔV smaller ($V_2 - V_1 < V_1 - V_0$) acceleration is lower than in 1	$S_2 = A_2$ $S_2 > S_1$
3	$W > R_3$ $R_3 > R_2$	Velocity increases from V_2 to V_3 ΔV smaller ($V_3 - V_2 < V_2 - V_1$)	$S_3 = A_3$ $S_3 > S_2$

Time Interval	Forces Involved	Changes in Velocity and Acceleration	Displacement
		Acceleration is lower than that in 2	
4	$W = R_4$	Velocity stays constant. $\Delta V = 0$. i.ee $V_3 = V_4 = \text{terminal velocity}$. Acceleration is 0	$S_4 = A_4$ $S_4 > S_3$

Small dense objects (eg steel balls) $\Rightarrow$ falls through air at the same acceleration and hit the ground at the same time.

- this is because they experience lower air resistance

In comparison $\Rightarrow$ piece of paper $\Rightarrow$ is light and has larger surface area

- experiences greater air resistance. Paper falls at lower acceleration

Objects experience higher air resistance when speed increases.

Factors Affecting Air Resistance

- Speed
- Shape
- Density of air

When air resistance $\Rightarrow$ acting against object $\Rightarrow$ equals its weight, object starts to travel at a constant speed known as **terminal velocity**

$\rightarrow$ means object has 0 acceleration

If object travels through short distance $\Rightarrow$ it may not reach terminal velocity before it hits the ground.

Skydiver A - spread eagle position (larger SA)

Skydiver B - head first position (smaller SA)

Skydiver A experience higher AR than skydiver B

Resultant force acting downwards on skydivers are their respective weights minus the respective air resistances acting upward on them.

Since weights of both skydivers are the same $\rightarrow$ resultant force acting downwards on skydiver A will be smaller than that acting on skydiver B.

Smaller resultant force means smaller acceleration for the same mass, thus smaller rate of increase in velocity. Hence, skydiver A is above skydiver B.

Answering Techniques

- Describe the motion when resultant force is acting on it in the opposite direction to the motion
- Object slows down in the direction of motion.
- Always state in which direction the force acts and between the 2 bodies.
- Explanation for increasing air resistance with increasing speed:
- The increase in air resistance will reduce the net force acting on the object. Thus, the acceleration of the object will decrease
- Always state the effect on the net force of the object. ALWAYS
- Always describe every factor from the equation;
- Example: for $F = Ma$, explain every factor (force, mass, acceleration) when using this formula

